



Unit 14 visual

A Inspection

MIDLAND COGEN

Outage Start Date: 20 Aug 2025

ESN/SY: 810893 | SY0048230

Oracle Project ID: A-1960052 | EV-185177 | EVP-555614

Report Issued: 20 Oct 2025

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OUTAGE DETAILS

1 Summary

1.1 Site Personnel

Name	Category	Role
Justin Clark	FieldCore	TFA

GAS TURBINE (810893 | SY0048230)

1 Summary

1.1 Timers and Counters

Technology	Gas Turbine
Total Starts	4982
Operating Hours	196258.9
Equivalent Operating Hours	295898.9

1.2 Gas Turbine Executive Summary

On August 21, 2025, a Visual Inspection was performed on Unit 14. During this inspection, Unit 14 was found to be in overall fair condition. The following are the results of the inspection:

A.) Tiles: Some yellow deposits were found on the surface of the tiles, as well as some minor coating loss.

B.) Secondary Air Nozzles: Medium erosion was found at the tips and the walls of the secondary air nozzles. One crack was found in the end of one and was blended out during this inspection.

C.) CIL: CIL was found with some erosion near secondary air nozzles and some erosion on the CIL near the secondary air nozzles.

D.) CIL Overlapping Ring: The overlapping ring was found in overall acceptable condition.

1.2 Gas Turbine Executive Summary

E.) HGC Collar: HGC collar was found with some minor erosion.

F.) Pulsation Probe Holder: The pulsation probe holder was found to be in acceptable condition.

G.) HGC: HGC was found with some coating loss. It is in overall acceptable condition.

H.) HGC Split Line Flange: HGC flange was found in acceptable condition.

I.) Inlet C-Segments: The c-segments were found with medium erosion.

J.) Row One Blades and Vanes: Row 1 blades and vanes were found with erosion on both leading and trailing edges.

K.) Row Two Blades and Vanes: Row 2 blades and vanes had minor erosion.

L.) Entry Segments: Entry segments were found with medium erosion.

M.) Burner: Burner was found with yellow deposits. Nozzle and swirler was in acceptable condition. A leak was found at one of the pipes on the cover that was repaired by the customer.

1.3 Work Scope

Outage Type: Planned Outage

Turbine
Hot Gas Path Inspection

2 Turbine

2.1 Upper Combustor (UCCI)

2.1.1 Strainer Basket

Part Description:

Strainer basket was found in acceptable condition.



IMG 1981

2.1.1 Strainer Basket



IMG 1987



IMG 1979

2.1.1 Strainer Basket



IMG 1983

2.1.2 Lower Cone

Part Description:

Lower cone was found in acceptable condition.

2.1.2 Lower Cone



IMG 1995

2.1.3 Burner

Part Description:

Burner was found with yellow deposits. Nozzle and swirler was in acceptable condition. A leak was found at one of the pipes on the cover that was repaired by the customer.

2.1.3 Burner



IMG 1968



IMG 1976

2.1.3 Burner



IMG 1967

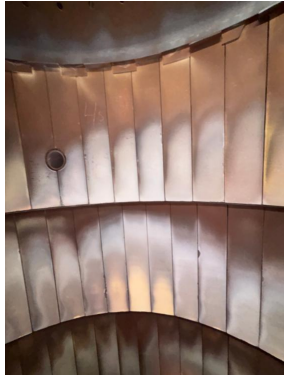


IMG 1970

2.1.4 Tile Carrier

Part Description:

Some yellow deposits were found on the surface of the tiles, as well as some minor coating loss.



IMG 1996



IMG 1997



IMG 2003



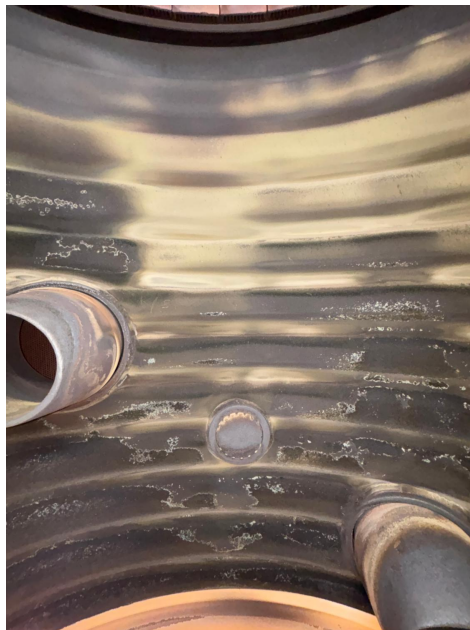
IMG 2004

2.2 Combustor Inner Liner (CIL)/(LCI)

2.2.1 Inner Liner

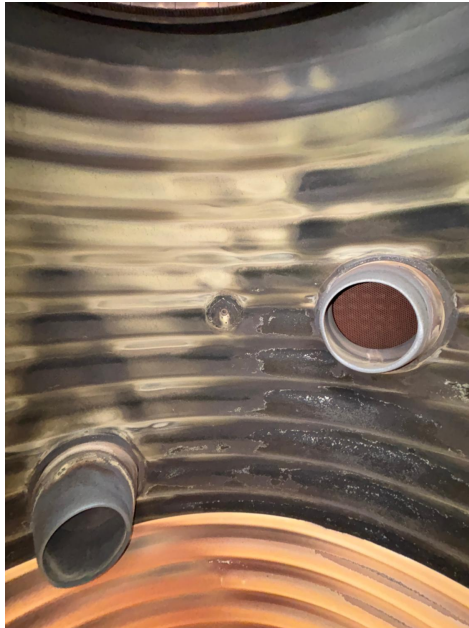
Part Description:

CIL was found with some erosion near secondary air nozzles and some erosion on the CIL near the secondary air nozzles.



IMG 2021

2.2.1 Inner Liner



IMG 2022



IMG 2025

2.2.1 Inner Liner



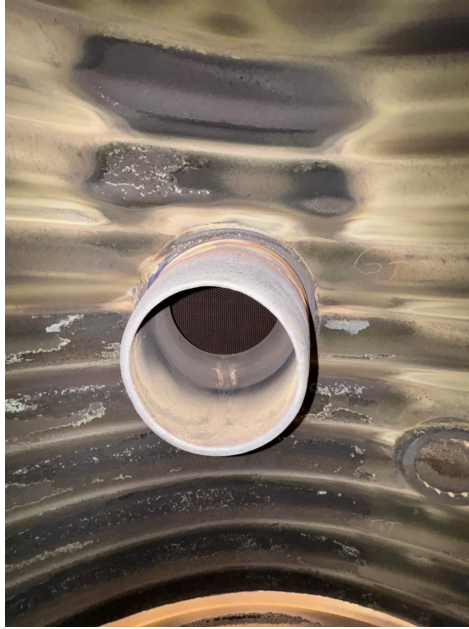
IMG 2026

2.2.2 Secondary Air Nozzles

Part Description:

Medium erosion was found at the tips and the walls of the secondary air nozzles. One crack was found in the end of one and was blended out during inspection.

2.2.2 Secondary Air Nozzles



IMG 2009

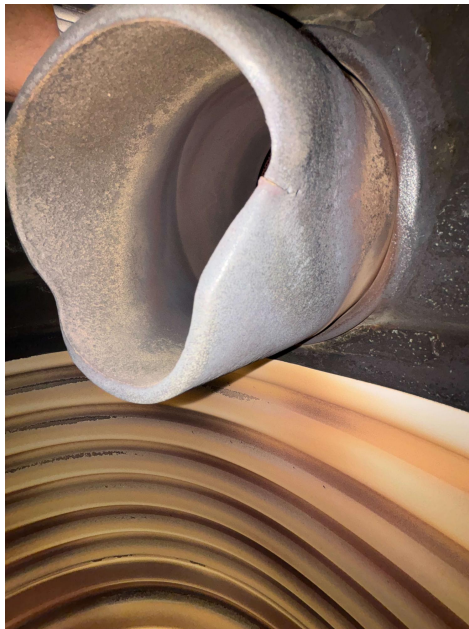


IMG 2061

2.2.2 Secondary Air Nozzles



IMG 2018



IMG 2015

2.2.3 Pulsation Probe Holder

Part Description:

Pulsation probe holder was found in acceptable condition.



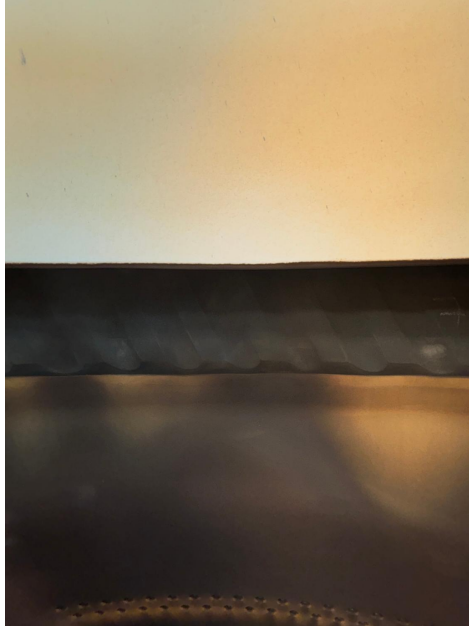
IMG 2006

2.3 Hot Gas Casing (HGC)

2.3.1 HGC collar

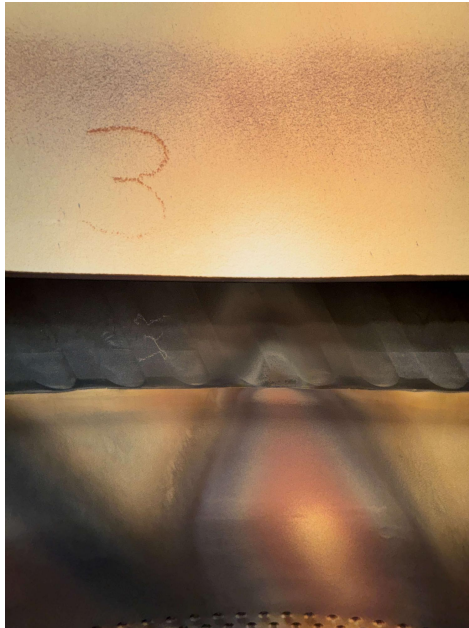
Part Description:

HGC collar was found with some minor erosion.

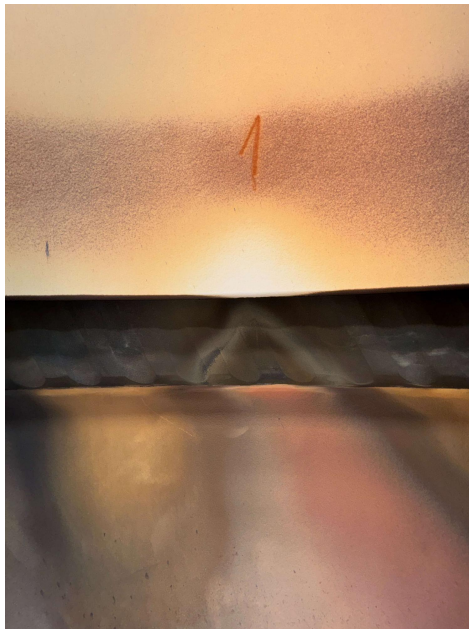


IMG 2028

2.3.1 HGC collar

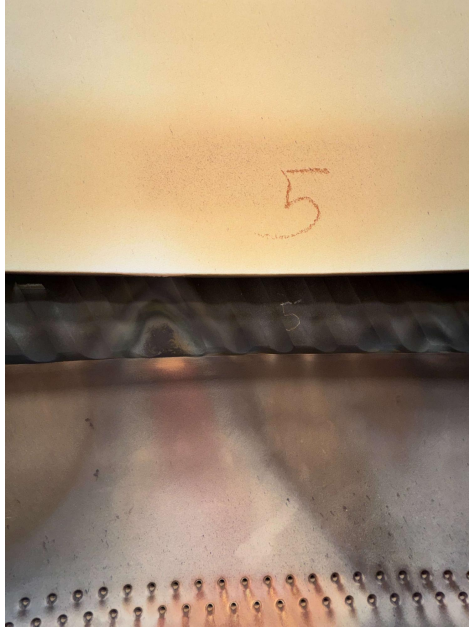


IMG 2029



IMG 2030

2.3.1 HGC collar



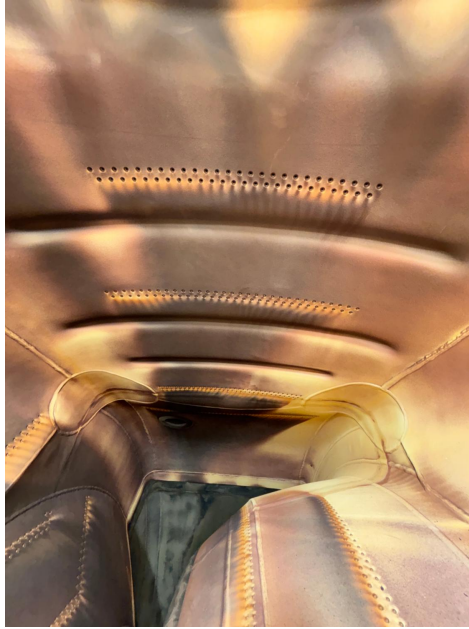
IMG 2027

2.3.2 HGC Upper and Lower

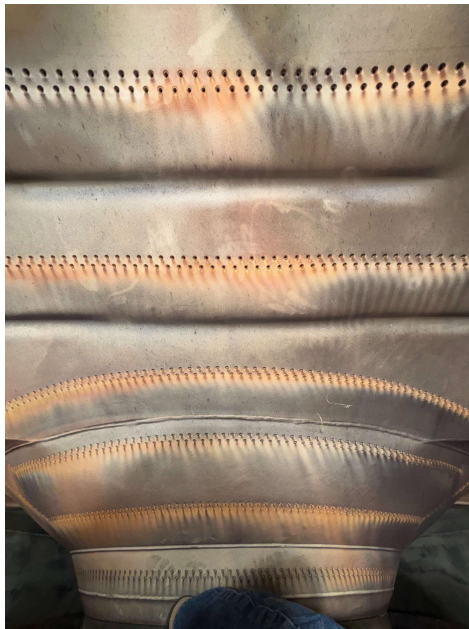
Part Description:

HGC was found with some coating loss. It is in overall acceptable condition.

2.3.2 HGC Upper and Lower



IMG 2031



IMG 2034

2.3.2 HGC Upper and Lower



IMG 2032



IMG 2033

2.3.3 HGC Flow Strut Separator

Part Description:

HGC flow strut separator was found in acceptable condition.



IMG 2053

2.3.3 HGC Flow Strut Separator



IMG 2043

2.4 Combustion Section

2.4.1 Entry Segments

Part Description:

Entry segments were found with medium erosion.

2.4.1 Entry Segments



IMG 2047



IMG 2048

2.4.2 C Segments

Part Description:

The C-Segments were found with medium erosion.

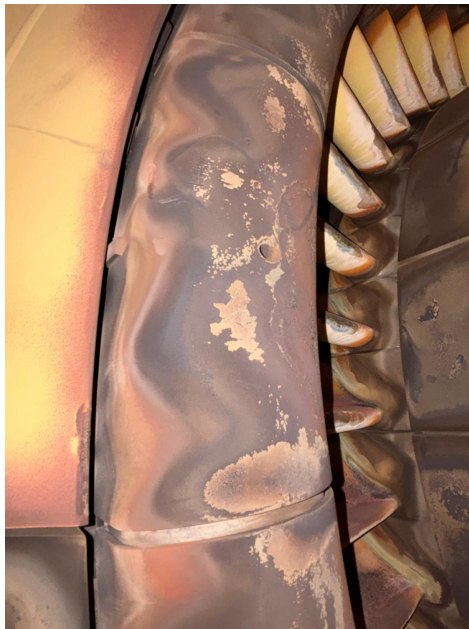


IMG 2037

2.4.2 C Segments



IMG 2054



IMG 2055

2.4.2 C Segments



IMG 2036

2.4.3 Vane Row 1

Part Description:

Row 1 vanes were found with erosion on both leading and trailing edges.

2.4.3 Vane Row 1



IMG 2038



IMG 2039

2.4.3 Vane Row 1



IMG 2040



IMG 2049

2.4.4 Blade Row 1

Part Description:

Row 1 blades were found with erosion on both leading and trailing edges.



2508210002

2.4.4 Blade Row 1



2508210003



IMG 2057

2.4.5 Blade and vane Row 2, 3, and 4

Part Description:

Row 2 blades and vanes had minor erosion. Rows 3 and 4 were founded in acceptable condition.



2508210004

2.4.5 Blade and vane Row 2, 3, and 4



2508210006



2508210005

2.4.5 Blade and vane Row 2, 3, and 4



2508210007



2508210008

2.4.5 Blade and vane Row 2, 3, and 4



2508210009

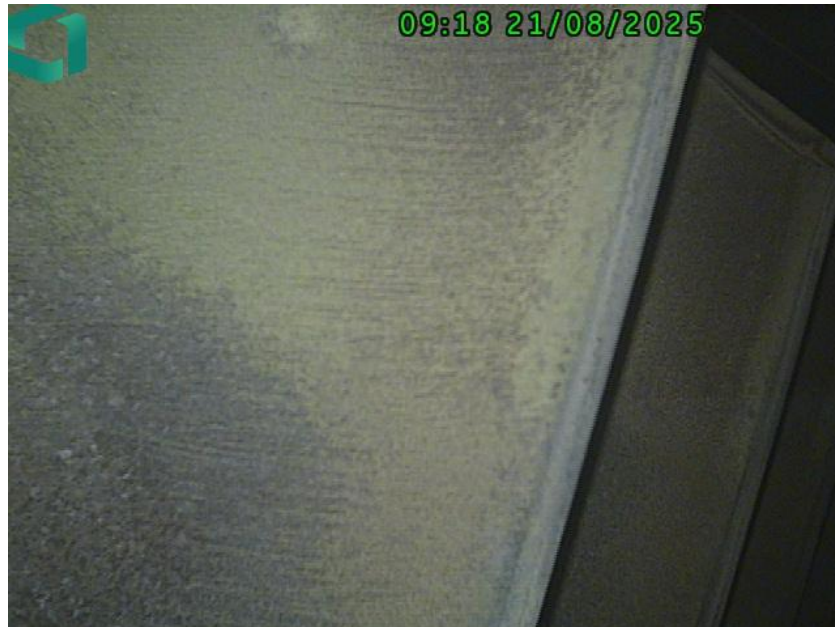


2508210010

2.4.5 Blade and vane Row 2, 3, and 4



2508210012



2508210013

2.4.5 Blade and vane Row 2, 3, and 4



2508210014



2508210015

2.4.5 Blade and vane Row 2, 3, and 4



2508210017



2508210018

2.4.5 Blade and vane Row 2, 3, and 4



2508210019

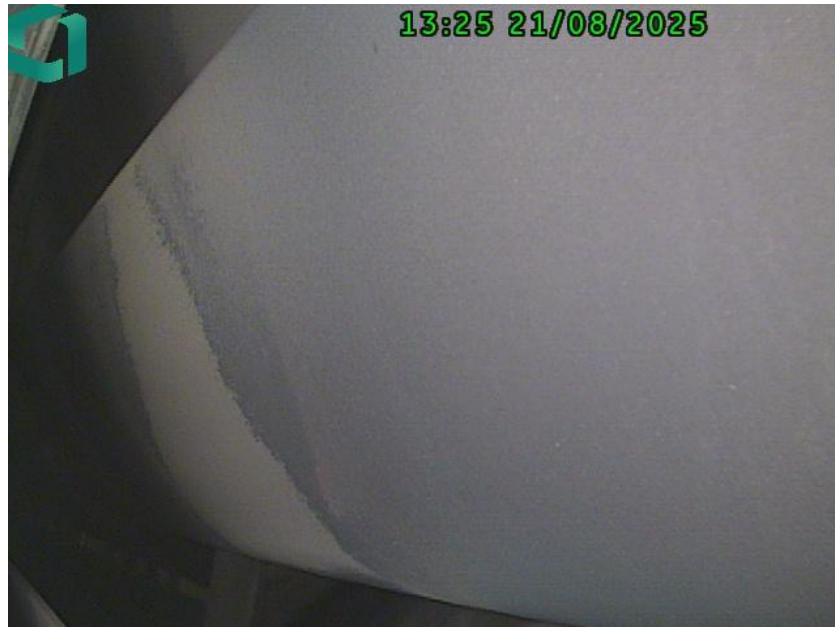


2508210020

2.4.5 Blade and vane Row 2, 3, and 4

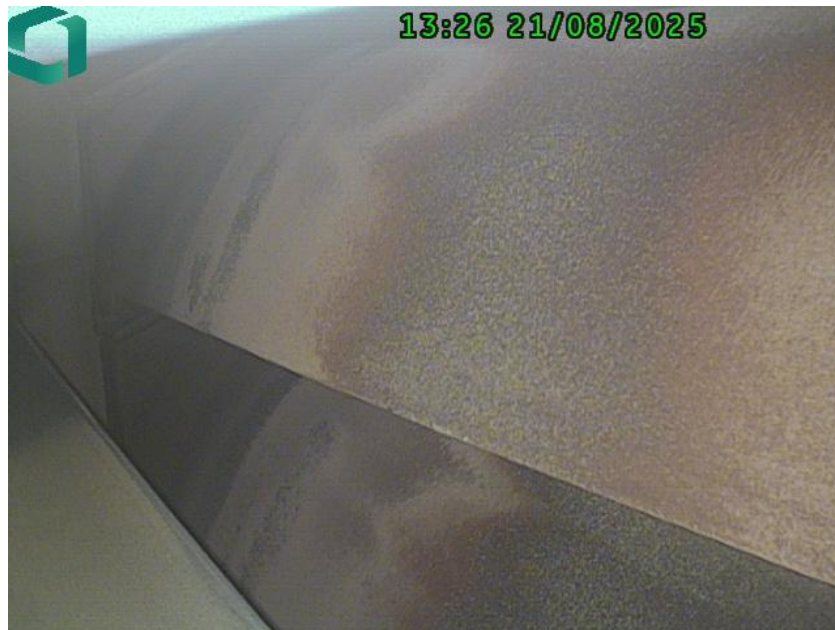


2508210021



2508210022

2.4.5 Blade and vane Row 2, 3, and 4



2508210024

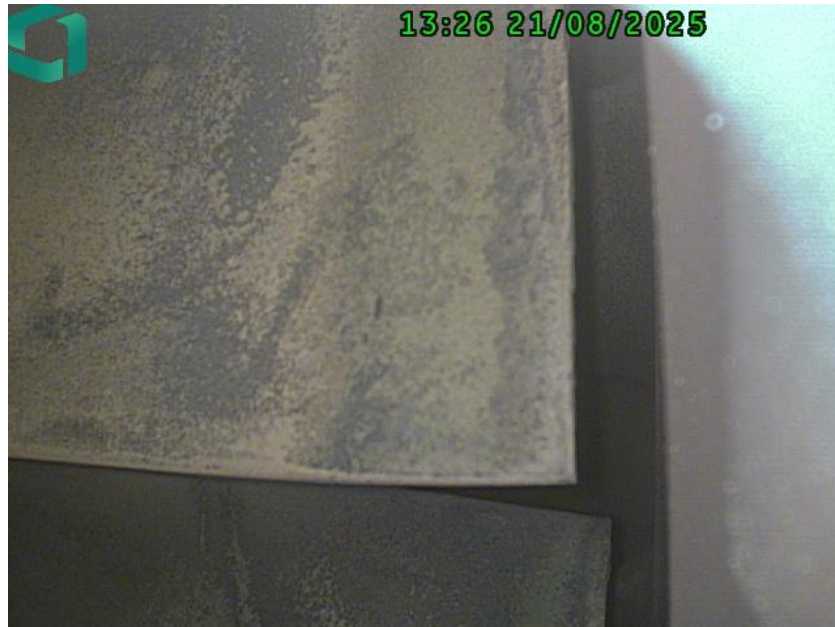


2508210025

2.4.5 Blade and vane Row 2, 3, and 4



2508210011



2508210026

2.4.6 Blade and Vane Row 5

Part Description:

Blade and Vane row 5 was found in acceptable condition.



IMG 2070

2.4.6 Blade and Vane Row 5



IMG 2071

2.5 Turbine Section

2.5.1 Exhaust Gas Housing and Exhaust Diffusor

Part Description:

Some cracks were found at the H brackets as well as the diffusor.

Recommendation Description:

Repair at next C inspection.

2.5.1 Exhaust Gas Housing and Exhaust Diffusor



IMG 2065



IMG 2064

2.5.1 Exhaust Gas Housing and Exhaust Diffusor



IMG 2067



IMG 2066

2.5.1 Exhaust Gas Housing and Exhaust Diffusor



IMG 2068